

ROBOTICS AND
AUTOMATION

Project Proposal



Detachable Tool-Changing Mechanism for Unitree G1

GROUP 1



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Introduction

01

Imagine a system where, just like Captain Hook's collection of tools, a robotic hand can quickly swap between different end-effectors

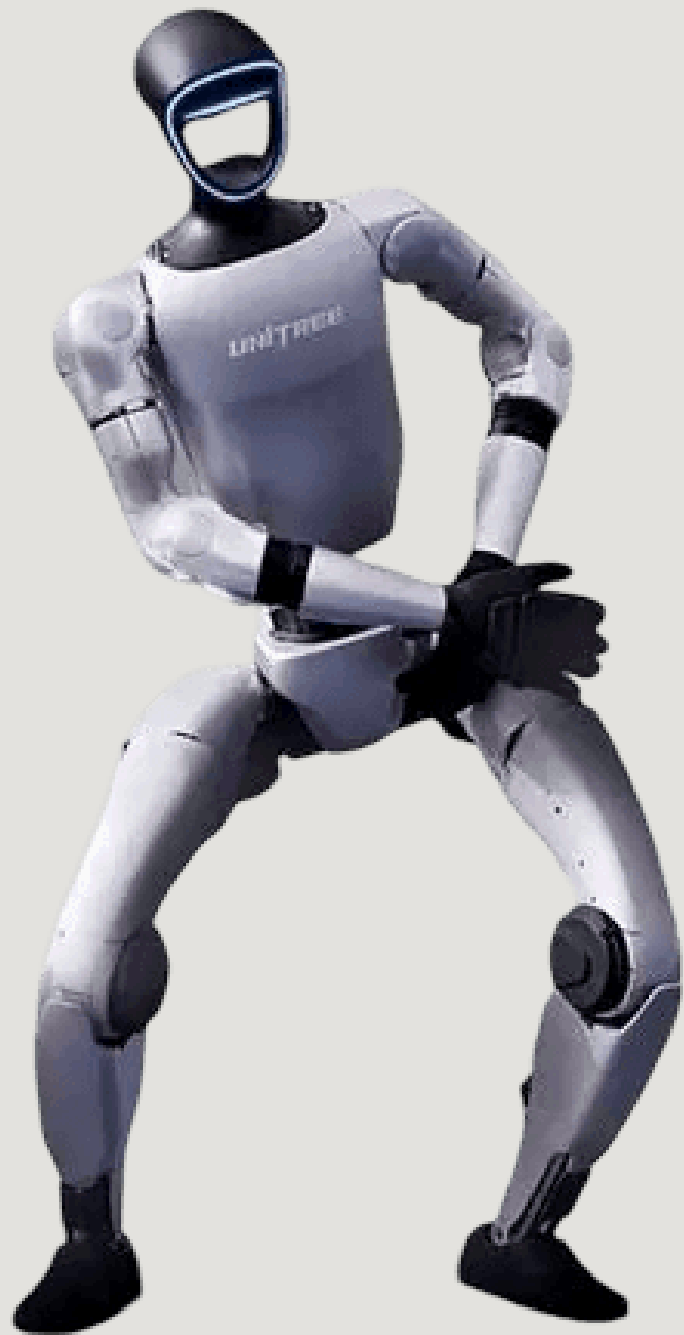
Each tool easily attaches and activates within seconds similar to the tool changers used in CNC machines.

This flexibility enables the robot to perform various tasks without needing to redesign or rebuild its hand.



About the Project

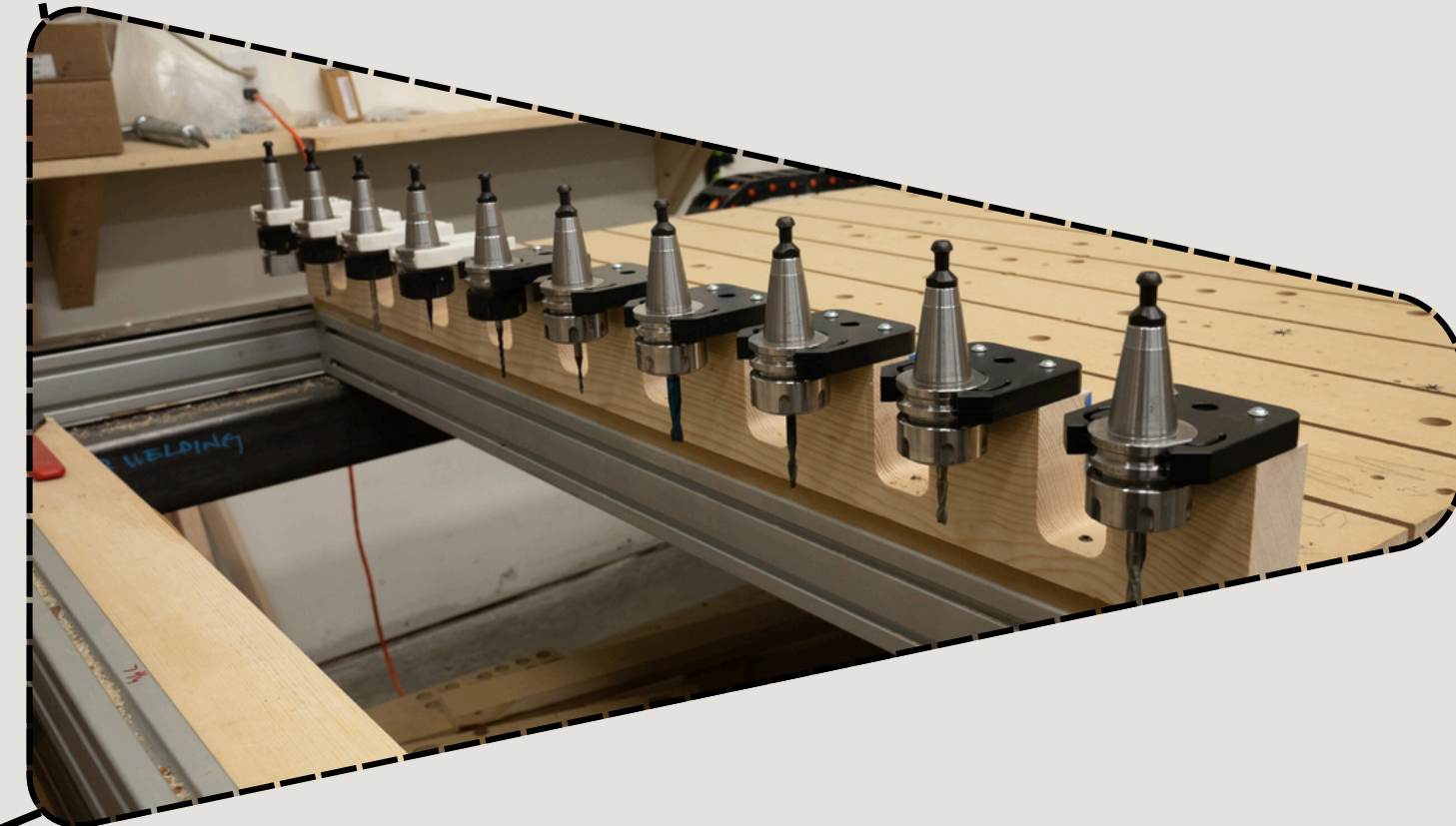
02



This project focuses on developing a detachable tool-changing mechanism for the Unitree G1 humanoid robot.

It allows for rapid switching between multiple end-effectors, including a drill, suction tool, and grapple

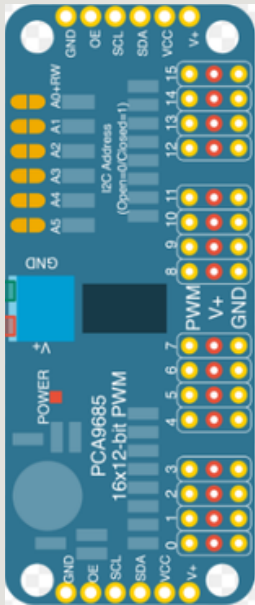
The robot will automatically plan a path to the docking station, changes the attached tool based on the user or task demand and activate the corresponding control mode.



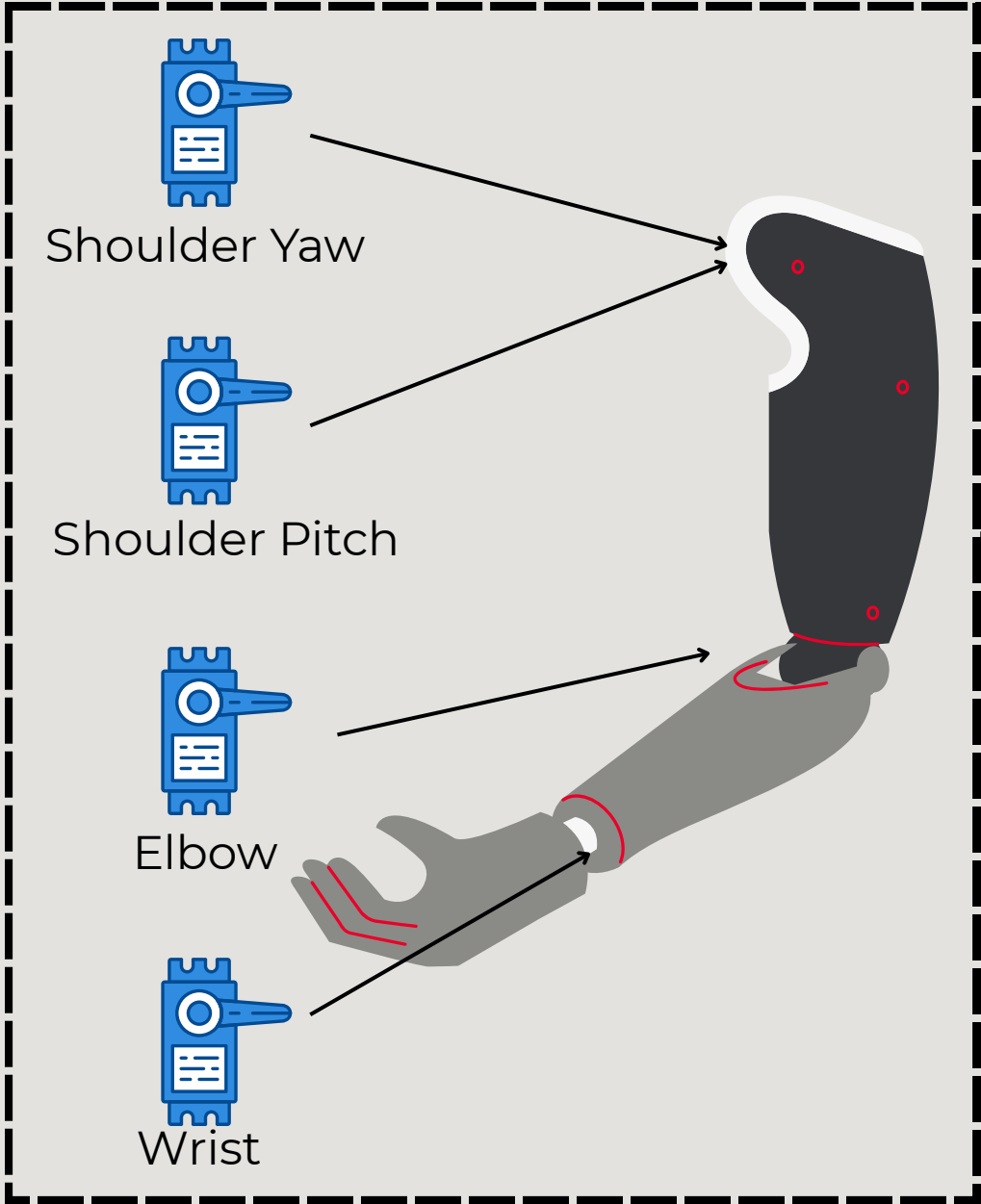
System Diagram



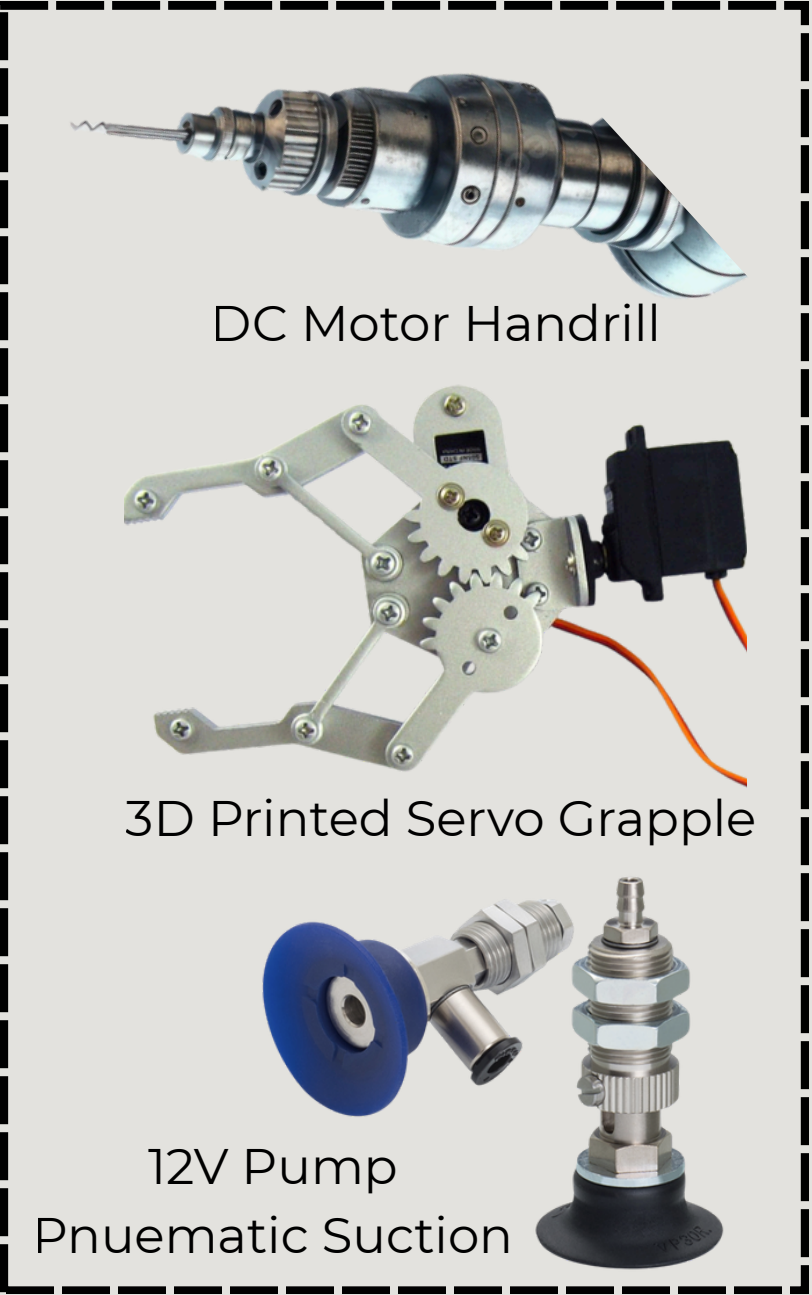
MCU



PWM Servo Driver



Special Coupling Mechanism ➔



Wrist Tools Exchanger

Main Tasks

04

PRELIMINARY DESIGN

2D Design
Diagrams
Concepts

DETAILED DESIGN

3D CAD
MATLAB Simulation
Electronic Testing

FABRICATION

Assembly
Wiring
Testing

CONTROL

Coordinate control
Path Planning
Tasks Testing

Q&A

